

step3_data_modeling

November 8, 2025

```
[2]: #Import libraries and load data

import pandas as pd
import joblib

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import f1_score, accuracy_score
```

```
[3]: X = pd.read_csv("preprocess_data/X_temp.csv")
y = pd.read_csv("preprocess_data/y_temp.csv").squeeze()
preprocessor = joblib.load("preprocess_data/preprocessor.pkl")

print("X shape:", X.shape)
print("y shape:", y.shape)
print("Data and preprocessor loaded.")
```

```
X shape: (20900, 18)
y shape: (20900,)
Data and preprocessor loaded.
```

```
[4]: #Train/Validation split

X_train, X_val, y_train, y_val = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Train samples:", X_train.shape[0])
print("Validation samples:", X_val.shape[0])
```

```
Train samples: 16720
Validation samples: 4180
```

[5]: *#Baseline Random Forest*

```
rf_base = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('model', RandomForestClassifier(random_state=42))
])

rf_base.fit(X_train, y_train)

y_pred = rf_base.predict(X_val)

print("Baseline Accuracy:", accuracy_score(y_val, y_pred))
print("Baseline F1:", f1_score(y_val, y_pred))
```

Baseline Accuracy: 0.815311004784689
Baseline F1: 0.7830241708825182

[6]: *#Define hyperparameter grid for tuning*

```
param_grid = {
    'model__n_estimators': [100, 200, 300],
    'model__max_depth': [None, 10, 20, 30],
    'model__min_samples_split': [2, 5, 10],
    'model__min_samples_leaf': [1, 2, 4],
    'model__max_features': ['sqrt', 'log2']
}

print("Parameter grid defined.")
```

Parameter grid defined.

[7]: *#Run Grid Search*

```
rf = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('model', RandomForestClassifier(random_state=42))
])

grid_search = GridSearchCV(
    rf,
    param_grid,
    cv=3,
    scoring='f1',
    n_jobs=-1,
    verbose=0
)
```

```
grid_search.fit(X_train, y_train)
```

```
[7]: GridSearchCV(cv=3,
                estimator=Pipeline(steps=[('preprocessor',
                                           ColumnTransformer(transformers=[('num',
Pipeline(steps=[('imputer',
                  SimpleImputer()),
                  ('scaler',
                    StandardScaler()))]),
                                           ['id',
'social-security-number',
'house-number',
'age-
group',
'fnlwgt',
'education-num',
'capitalgain',
'capitalloss',
'hoursperweek']),
                                           ('cat',
Pipeline(steps=[('imputer',
                  SimpleImputer(s...
'occupation',
'relationship',
'race',
'sex',
'native-country-code',
'native-country'])])),
                ('model',
RandomForestClassifier(random_state=42))]),
        n_jobs=-1,
        param_grid={'model__max_depth': [None, 10, 20, 30],
                    'model__max_features': ['sqrt', 'log2'],
                    'model__min_samples_leaf': [1, 2, 4],
                    'model__min_samples_split': [2, 5, 10],
                    'model__n_estimators': [100, 200, 300]},
        scoring='f1')
```

```
[12]: print("Grid search completed.")
      print("Best Parameters:", grid_search.best_params_)
      print("Best CV F1 Score:", grid_search.best_score_)
```

Grid search completed.

Best Parameters: {'model__max_depth': 20, 'model__max_features': 'sqrt',
'model__min_samples_leaf': 1, 'model__min_samples_split': 2,
'model__n_estimators': 300}

Best CV F1 Score: 0.7920295236934362

```
[13]: #Save tuned model for Step 4  
  
joblib.dump(grid_search, "preprocess_data/grid_rf.pkl")  
print("Saved grid_rf.pkl for Step 4.")
```

Saved grid_rf.pkl for Step 4.